D.A.R.I.A. – Domain-Specific Agentic Research, Intelligence & Analysis System

Memory-Driven Hybrid Research, Reflection and Summarization using LangGraph, RAG, Web Search and Persistent Memory

Abstract

D.A.R.I.A. is a domain-specific agentic research system designed to address limitations of traditional Retrieval-Augmented Generation (RAG) systems. The platform integrates persistent memory, research planning, hybrid retrieval, evidence curation, reflection loops, and structured summarization into a unified LangGraph workflow. Rather than producing immediate responses from retrieved information, the system follows a multi-stage research process that resembles how a human researcher plans, gathers evidence, evaluates quality, revises strategy, and synthesizes findings. The result is an explainable, auditable, and continuously improving research workflow.

1. Introduction

Recent advances in Large Language Models have enabled powerful question-answering systems. However, most systems remain fundamentally reactive. Traditional RAG pipelines retrieve information and generate responses but lack long-term memory, self-evaluation, and adaptive research strategies. D.A.R.I.A. was developed to bridge this gap by introducing memory-driven planning, critique-based reflection, and persistent learning.

The project demonstrates how agentic AI principles can be applied to research workflows. By combining multiple specialized agents with state management and reflection mechanisms, the system transforms research from a one-shot retrieval process into an iterative and evidence-driven workflow.

2. Problem Statement

Conventional RAG systems face several challenges:

- No memory of previous research sessions.
- Limited transparency regarding reasoning and retrieval decisions.
- No mechanism for evaluating evidence quality.
- Inability to revise search strategies after failure.
- Lack of structured research planning.

D.A.R.I.A. addresses these challenges through persistent memory, planning agents, evidence curation, critic-driven evaluation, reflection loops, and structured reporting.

3. System Architecture

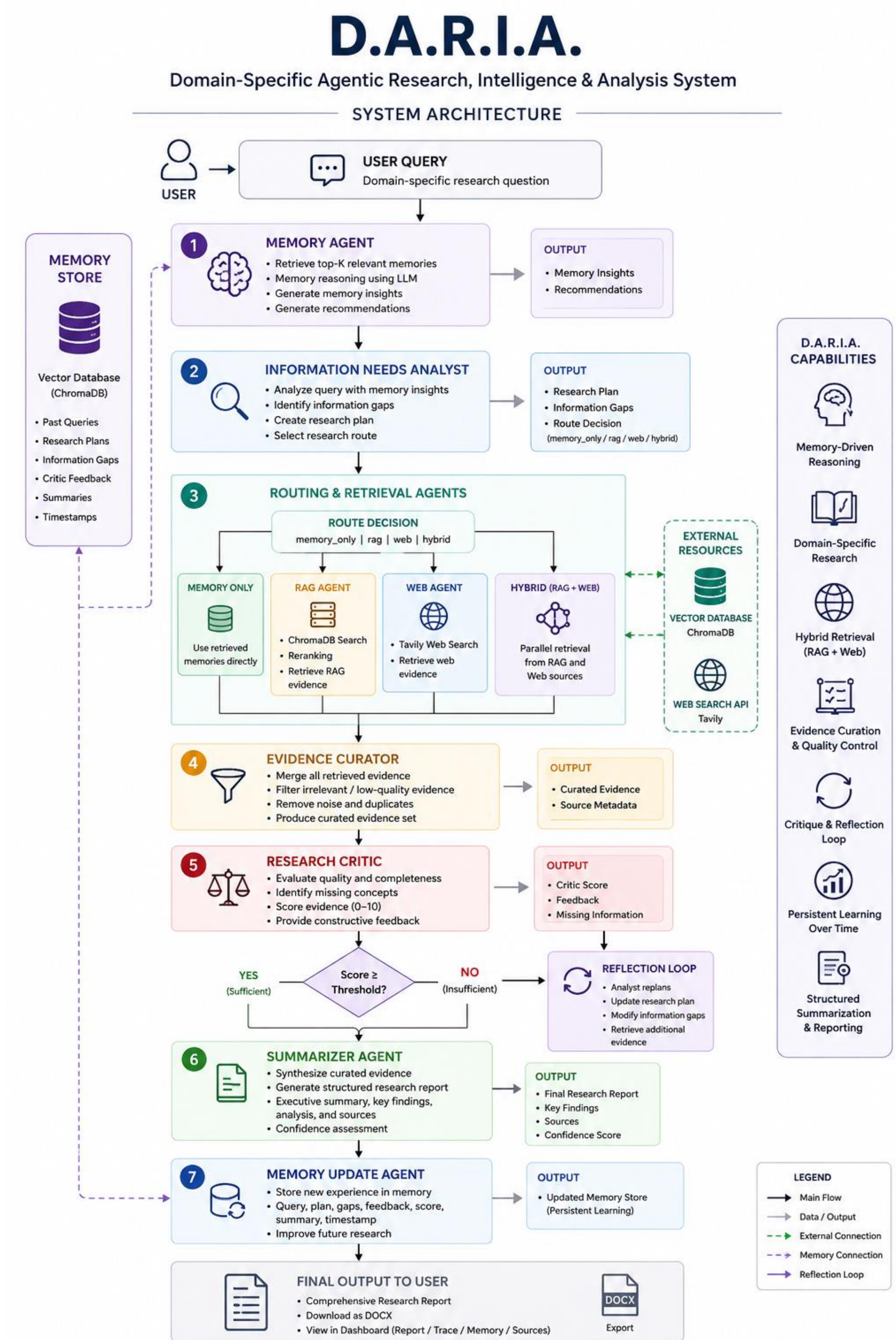

Figure 1. D.A.R.I.A. System Architecture

The architecture consists of five major layers:

Memory Layer:
Stores previous research experiences including plans, information gaps, critic feedback, scores, and summaries.

Planning Layer:
Analyzes user intent, identifies information gaps, and selects an optimal research route.

Retrieval Layer:
Performs domain-specific RAG retrieval, web search, or hybrid retrieval through parallel execution.

Evaluation Layer:
Filters evidence and evaluates research quality using a dedicated critic agent.

Learning Layer:
Stores research outcomes and feedback to improve future investigations.

4. LangGraph Workflow Design

D.A.R.I.A. is implemented using LangGraph and a shared ResearchState object. Each node is responsible for a specialized task while state transitions are managed through conditional routing.

Workflow Characteristics:
• Conditional routing.
• Parallel execution.
• Reflection loops.
• State persistence.
• Memory integration.

Routing Modes:
1. memory_only
2. rag
3. web
4. hybrid

The hybrid route executes RAG and Web Agents concurrently and merges results before evaluation.

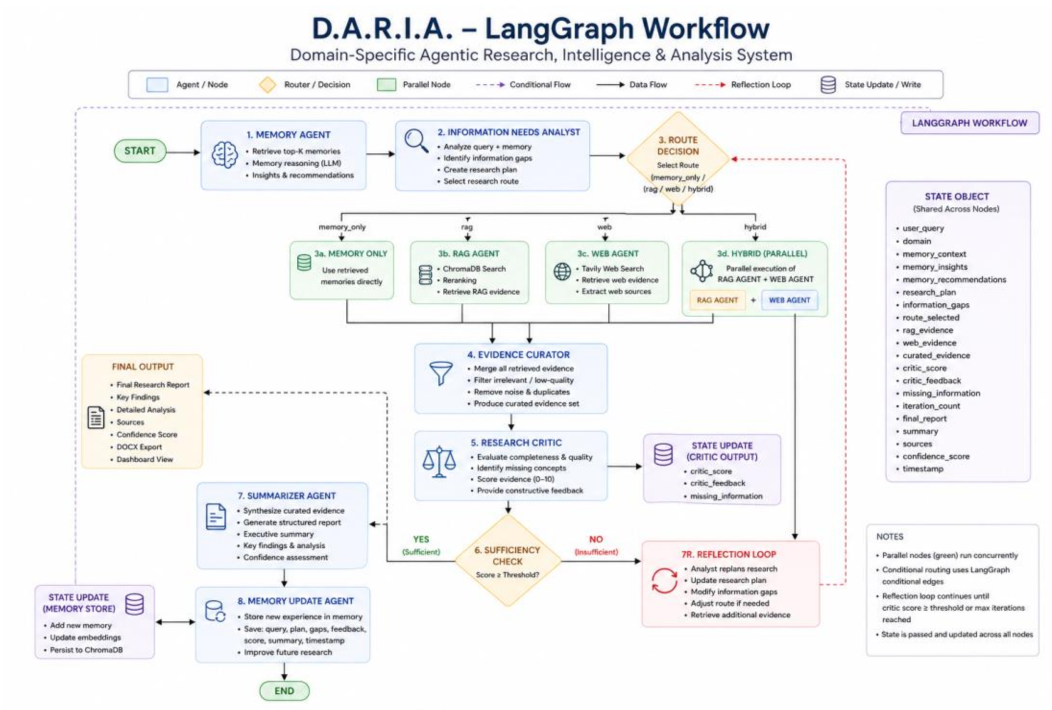

Figure 2. LangGraph Workflow and Conditional Routing

5. Agent Design

5.1 Memory Agent

The Memory Agent retrieves previous research sessions and generates memory insights and recommendations. Retrieved memories influence planning and help avoid repeating failed research strategies.

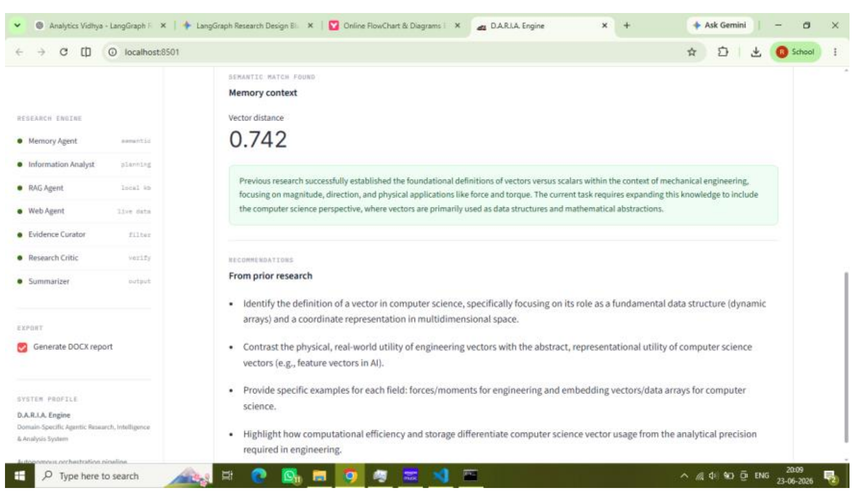

Figure 3: Memory Summary

5.2 Information Needs Analyst

The Information Needs Analyst converts user questions into structured research plans. It identifies missing concepts, determines information gaps, and selects retrieval strategies.

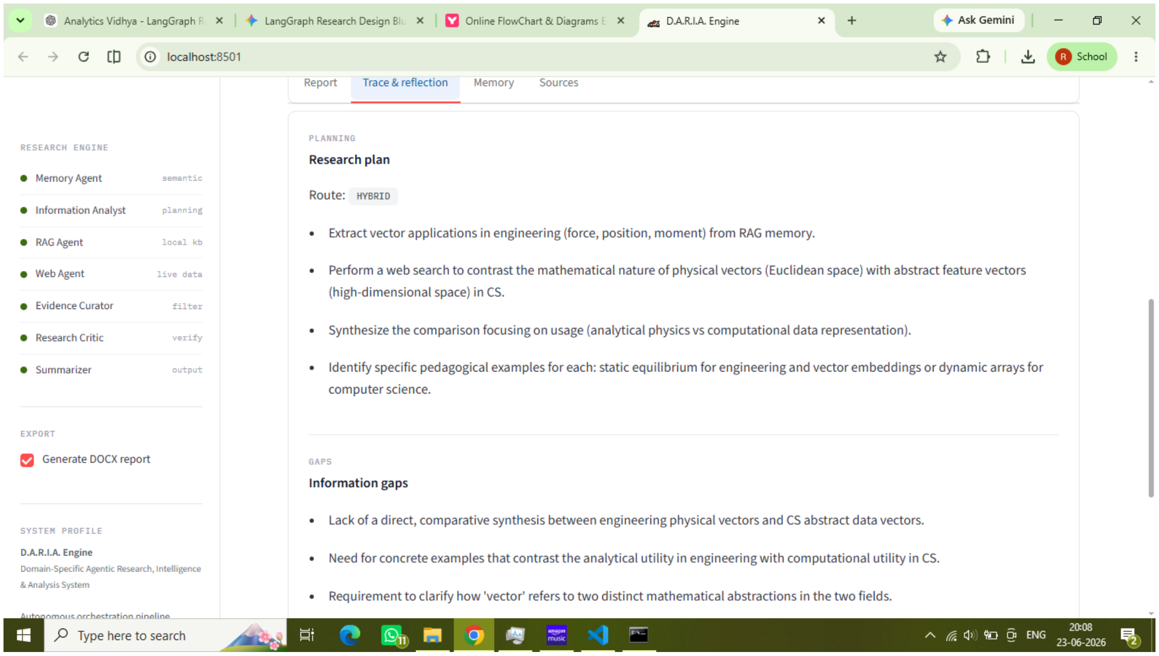

Figure 4: Research Planning

5.3 RAG Agent

The RAG Agent retrieves domain-specific knowledge from a ChromaDB vector store built from engineering mechanics resources. Semantic search and reranking improve evidence quality.

5.4 Web Agent

The Web Agent retrieves dynamic and recent information using Tavily search. It complements domain-specific retrieval when knowledge is unavailable within the corpus.

5.5 Hybrid Retrieval

Hybrid retrieval executes RAG and Web Agents concurrently. This improves coverage while maintaining domain relevance.

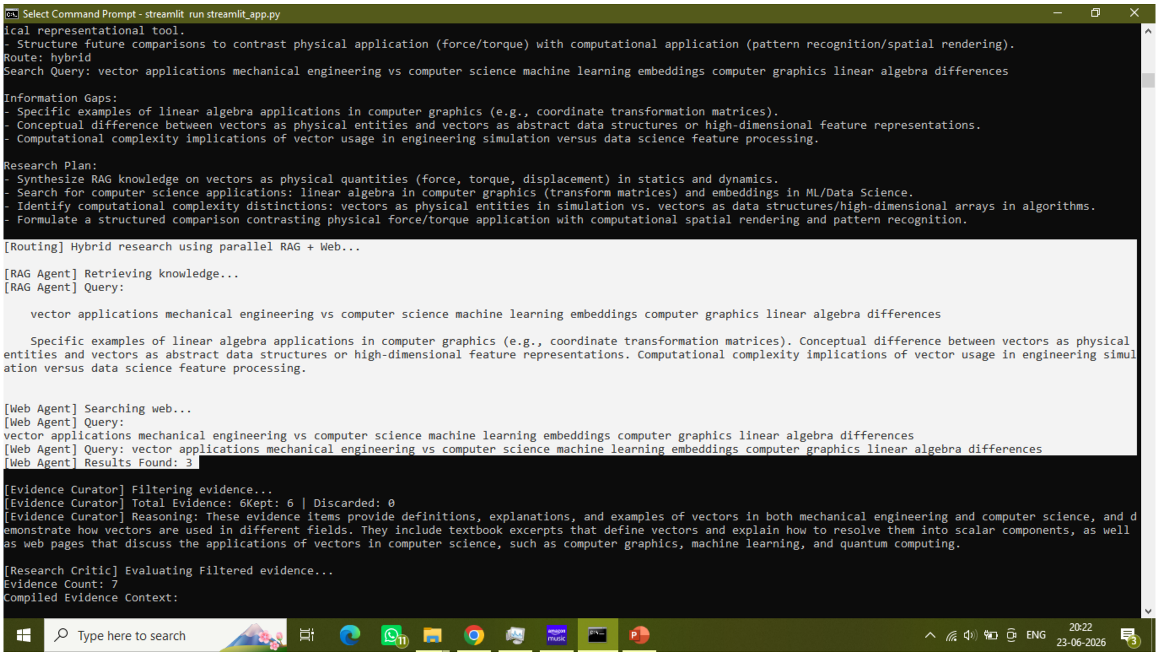

Figure 5: Hybrid Retrieval

5.6 Evidence Curator

The Evidence Curator merges and filters evidence before evaluation. Irrelevant, redundant, or weak evidence is discarded.

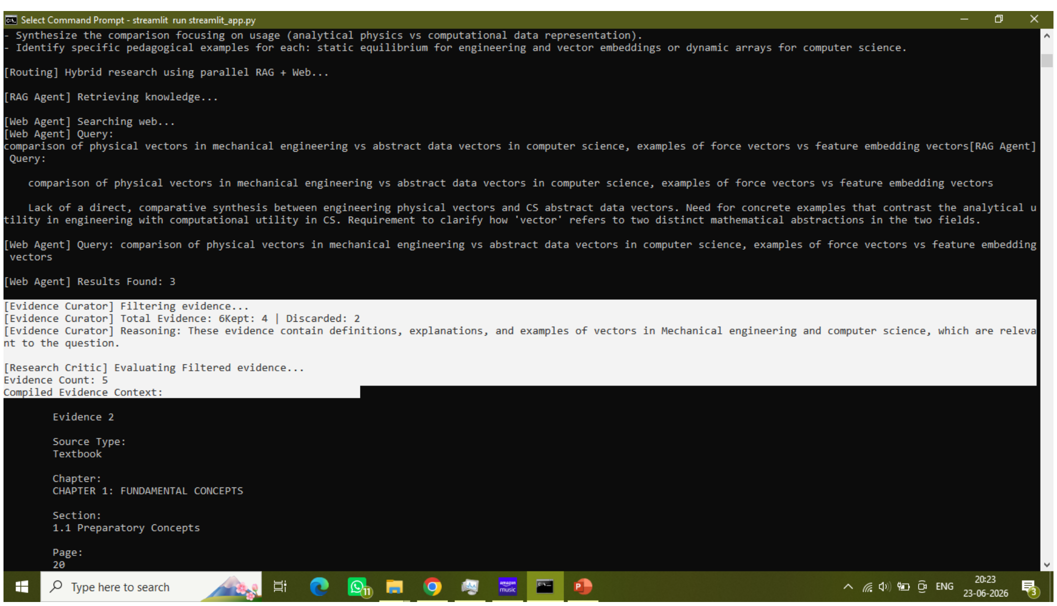

Figure 6: Evidence Curation

5.7 Research Critic

The Research Critic evaluates completeness, relevance, and evidence quality. It assigns scores and identifies deficiencies.

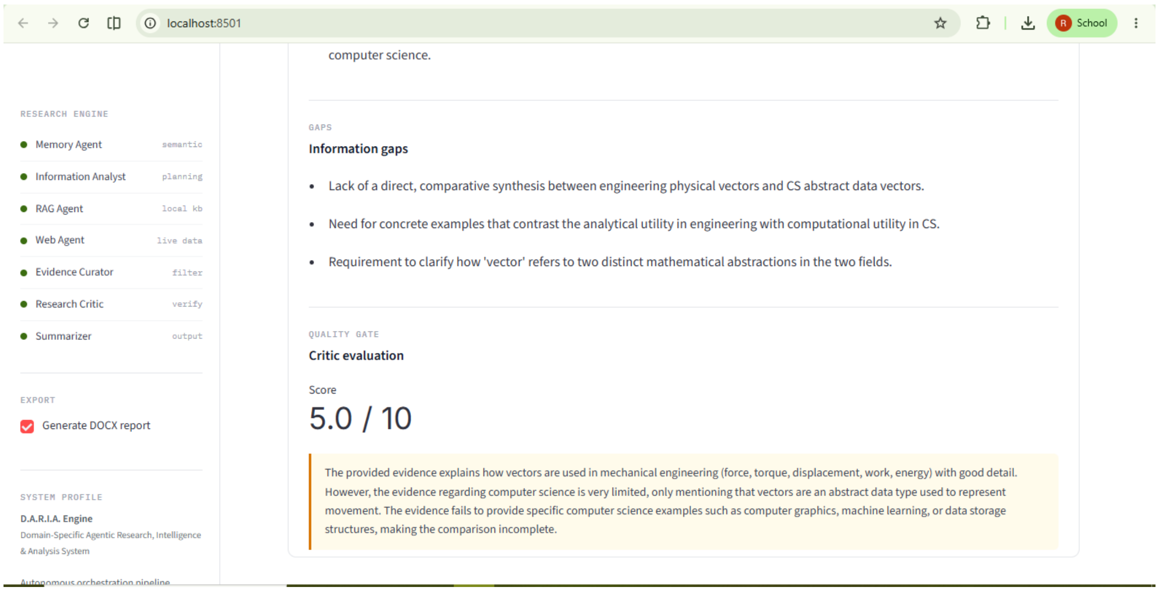

Figure 7: Research Critic

5.8 Summarizer Agent

The Summarizer synthesizes validated evidence into structured reports including executive summaries, findings, analysis, sources, and confidence assessments.

5.9 Memory Update Agent

The Memory Update Agent stores research experiences, enabling persistent learning across sessions.

6. Reflection-Based Research Improvement

A major innovation of D.A.R.I.A. is the reflection loop. After retrieval and curation, evidence is evaluated by the Research Critic. If evidence quality is insufficient, critic feedback is transformed into new information gaps and research plans.

This mechanism enables:
• Self-correction.

• Iterative improvement.
• Adaptive retrieval.
• Increased research completeness.

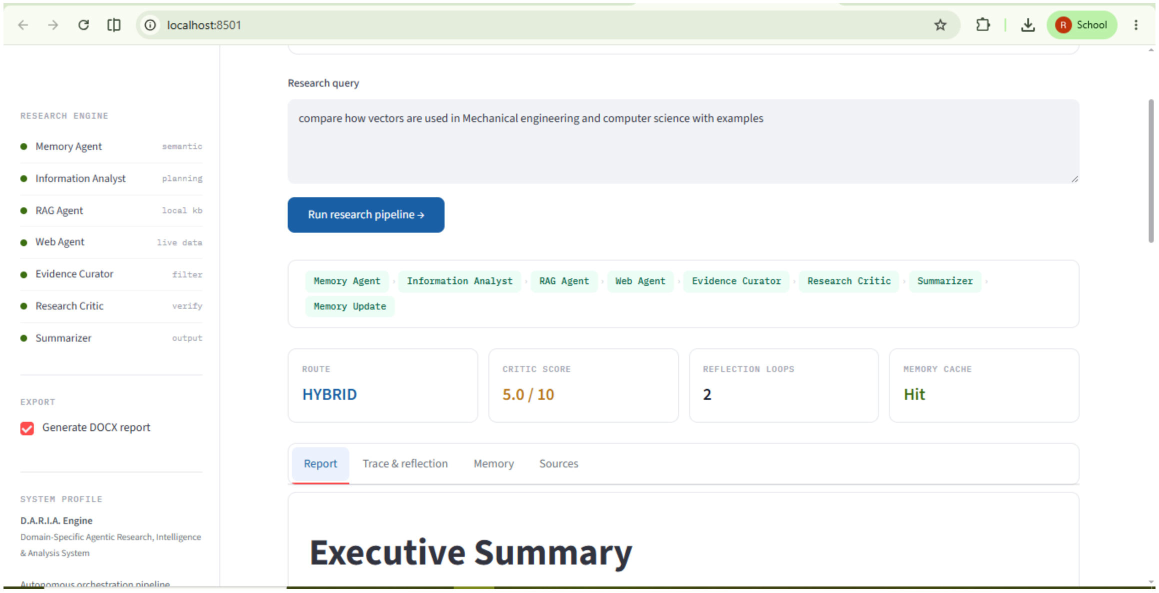

Figure 8. Reflection Loop

7. Report Generation and User Interface

The Streamlit interface exposes the entire research workflow. Users can inspect research plans, information gaps, critic feedback, memory insights, and sources.

The final report includes:
• Executive Summary
• Key Findings
• Detailed Analysis
• Sources
• Confidence Assessment

Reports can be exported as DOCX documents.

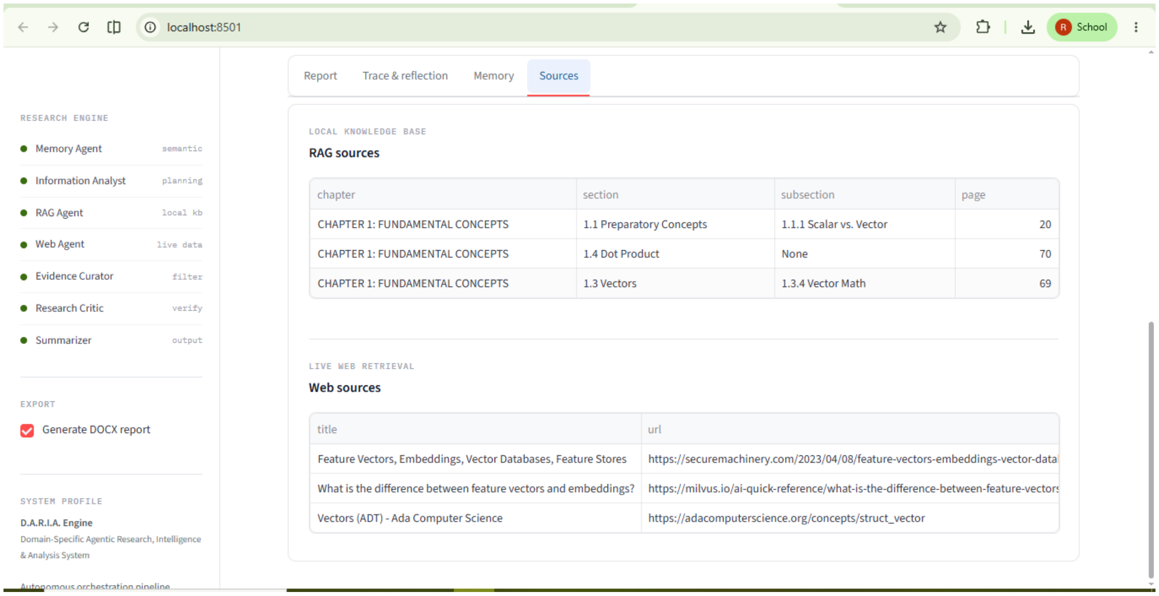

Figure 9: Sources tab

8. Technical Stack

Agent Framework: LangGraph
LLM Orchestration: LiteLLM
Research Model: Gemini 2.5 Flash
Vector Database: ChromaDB
Embeddings: BAAI BGE Small
Web Search: Tavily
Frontend: Streamlit
Document Export: python-docx
Knowledge Extraction: Docling

9. Key Innovations

9.1 Memory-Augmented Research

Traditional RAG systems operate without long-term memory. D.A.R.I.A. stores research plans, information gaps, critic feedback, and summaries. Future sessions benefit from accumulated experience.

9.2 Research Planning Layer

The Information Needs Analyst explicitly plans research before retrieval. This shifts the workflow from reactive retrieval to goal-driven investigation.

9.3 Corpus-Aware Routing

Different research problems require different retrieval strategies. D.A.R.I.A. dynamically selects memory, RAG, web, or hybrid routes.

9.4 Parallel Hybrid Retrieval

Concurrent execution of RAG and Web Agents improves coverage while reducing latency.

9.5 Evidence Curation

Evidence is filtered before summarization. This reduces hallucinations and improves answer quality.

9.6 Critique-Based Evaluation

The Research Critic acts as an independent evaluator that measures evidence quality and identifies deficiencies.

9.7 Reflection Loop

Rather than accepting imperfect evidence, the system replans and retries retrieval. This is one of the most agentic aspects of the architecture.

9.8 Persistent Learning

Research outcomes are stored for future use. The system becomes increasingly useful as additional research sessions accumulate.

9.9 Explainable Research

Research plans, gaps, scores, and feedback are exposed to users. This creates transparency and trust.

9.10 Structured Research Reporting

Instead of generating conversational responses, the system produces professional research reports with traceable sources and confidence assessments.

10. Results and Demonstration

Testing demonstrated successful operation of memory retrieval, hybrid retrieval, evidence curation, critic-driven reflection, structured summarization, and DOCX export.

Key outcomes:
• Successful memory integration.
• Dynamic routing.
• Parallel retrieval.
• Reflection-driven improvements.
• Explainable workflows.
• Professional report generation.

11. Limitations

Current limitations include:
• Single-domain corpus.
• Threshold-based memory retrieval.
• Fixed reflection iterations.
• Text-centric evidence processing.
• Limited source reliability assessment.

12. Future Work (D.A.R.I.A. V2)

Future enhancements include:
• Multi-domain corpora.
• Multimodal RAG.
• Memory relevance ranking.
• Adaptive retrieval strategies.
• Source credibility scoring.
• Autonomous task decomposition.
• Performance analytics dashboards.
• Agent benchmarking.

13. Conclusion

D.A.R.I.A. demonstrates how memory, planning, retrieval, critique, reflection, and summarization can be integrated into a unified research workflow. By combining LangGraph orchestration, domain-specific RAG, web search, evidence curation, and persistent memory, the system moves beyond traditional question-answering architectures toward explainable and continuously improving research systems. The project highlights the potential of agentic AI for engineering education, research support, and knowledge-intensive decision-making.